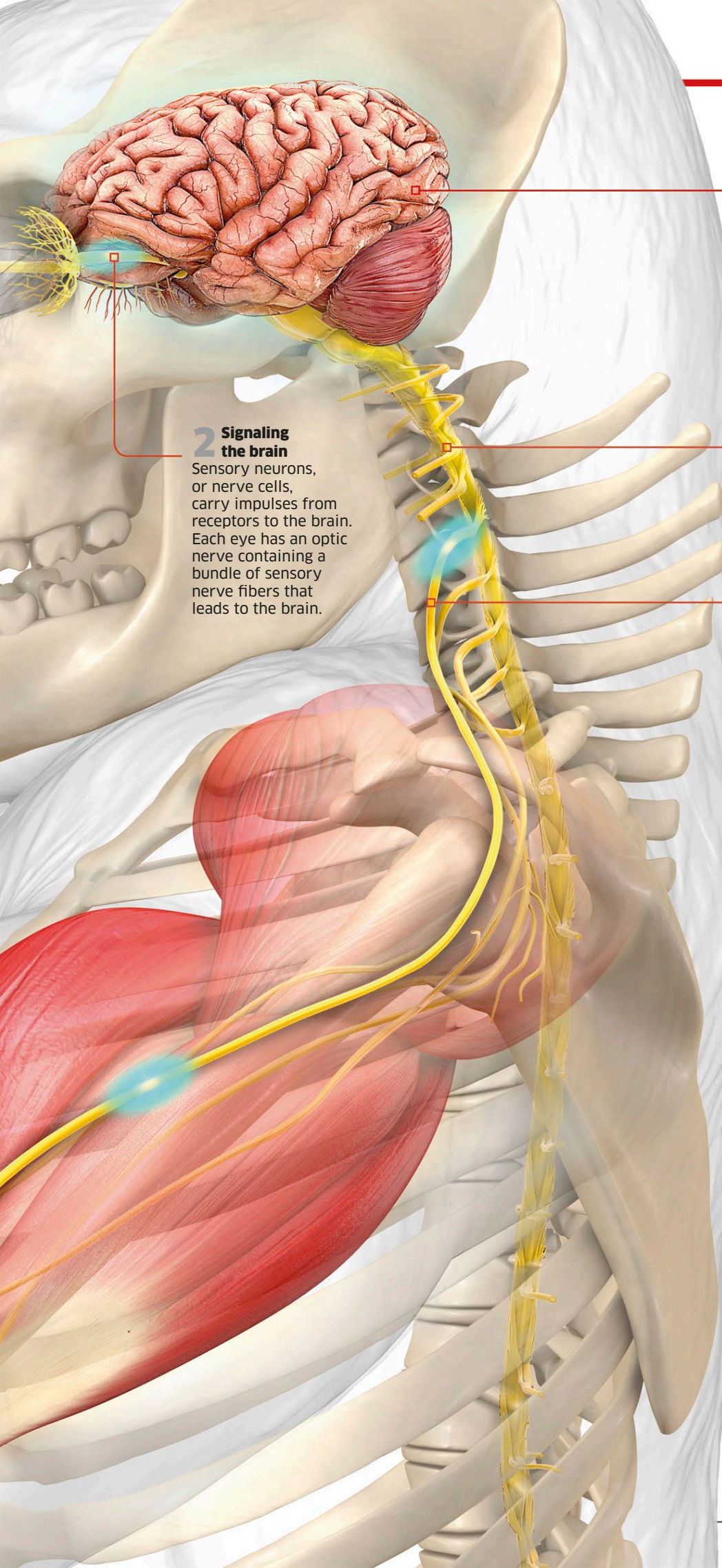
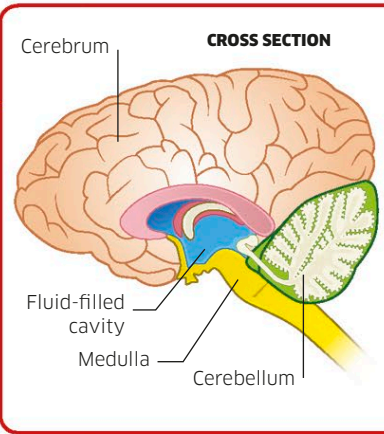




2 Signaling the brain

Sensory neurons, or nerve cells, carry impulses from receptors to the brain. Each eye has an optic nerve containing a bundle of sensory nerve fibers that leads to the brain.



3 Coordinating a response

The brain coordinates where impulses go in order to control the body's behavior. The cerebrum manages complex actions that demand intelligence, like peeling and breaking up food. More routine actions, such as walking, are controlled by the cerebellum, while the medulla effects internal functions, like breathing.

4 Traveling onward

Together with the brain, the spinal cord makes up the central nervous system. It works with the brain to pass signals around the body. Impulses traveling from the brain branch off from the spinal cord to motor neurons.

5 Signaling the muscles

Cells that carry impulses from the central nervous system to muscles are called motor neurons. Bundles of motor neuron fibers are grouped into nerves that run all the way from the spinal cord to the limb muscles.

Reflex actions

Some automatic responses, called reflex actions, do not involve the brain, such as when you recoil after touching something hot. In these instances, impulses travel from the sense organs to the spinal cord, where relay neurons pass the signal to the muscles. Bypassing the brain allows the impulses to reach the effectors and generate a response much more quickly.

2 Relay neuron

This passes nerve impulses from sensory to motor neurons. It can also pass signals up the spinal cord and to the brain.

3 Motor neuron

By transmitting nerve impulses from spinal cord to muscles, this triggers movement.

1 Sensory neuron

This carries nerve impulses from a sense organ into the spinal cord.

